

數位電路實驗 Lab2 team9

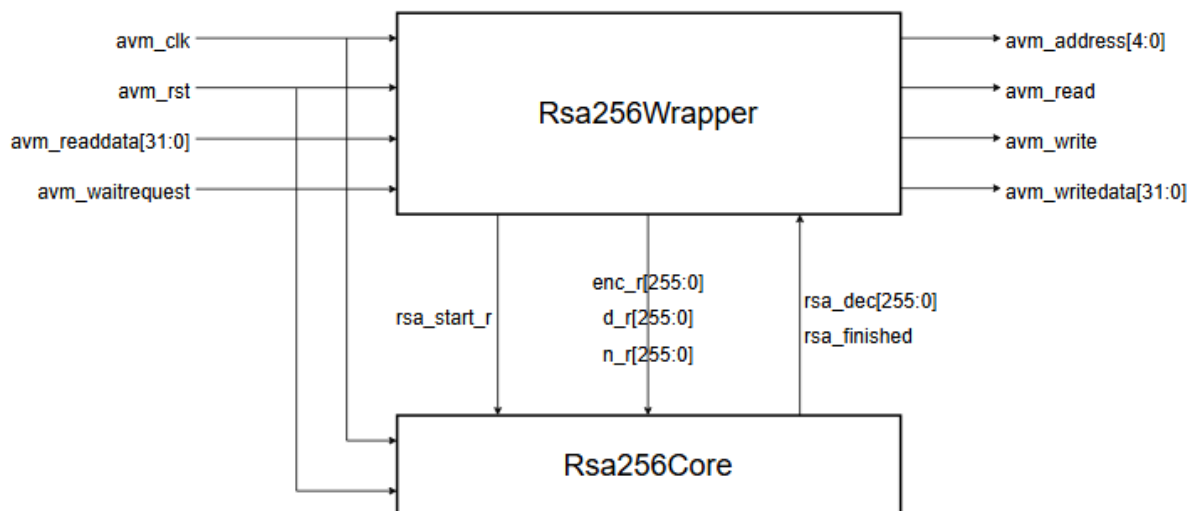
B12901005 郭力銘 B12901185 謝承恩 B12901191 彭子綸

1. File Structure

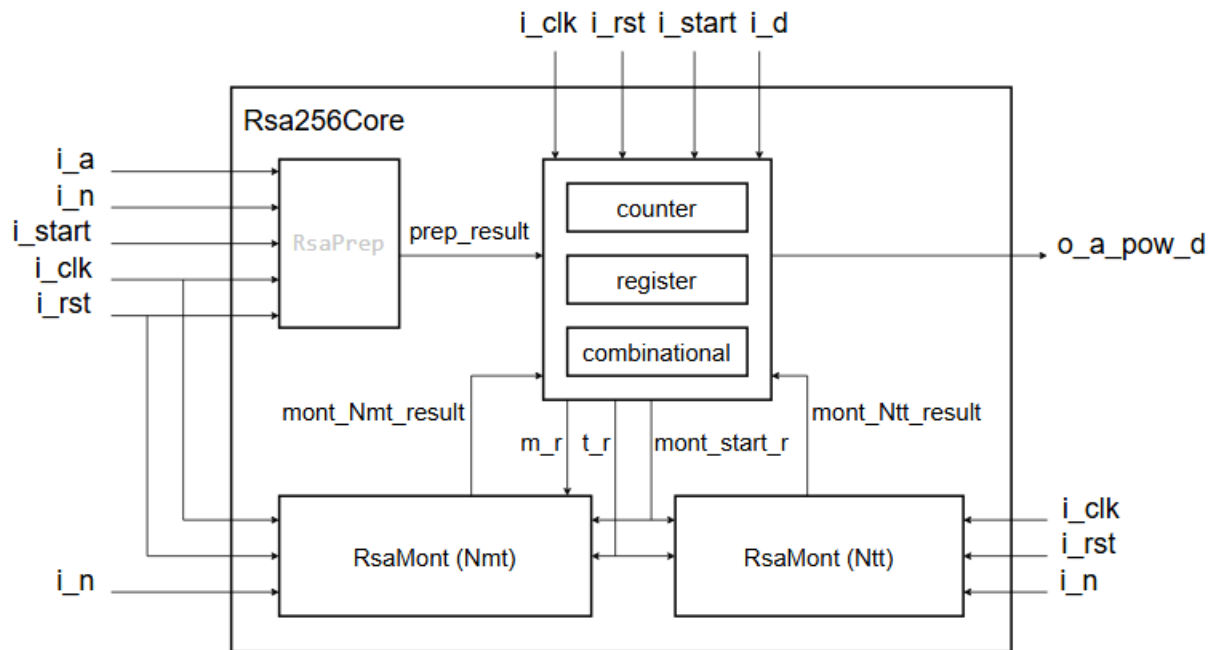
```
team09_lab2
|- team09_lab2_report
|- src
|  |- DE2_115
|     |- DE2_115.qsf
|     |- DE2_115.sdc
|     |- DE2_115.sv
|  |- Rsa256Core.sv
|  |- Rsa256Wrapper.sv
```

2. System Architecture

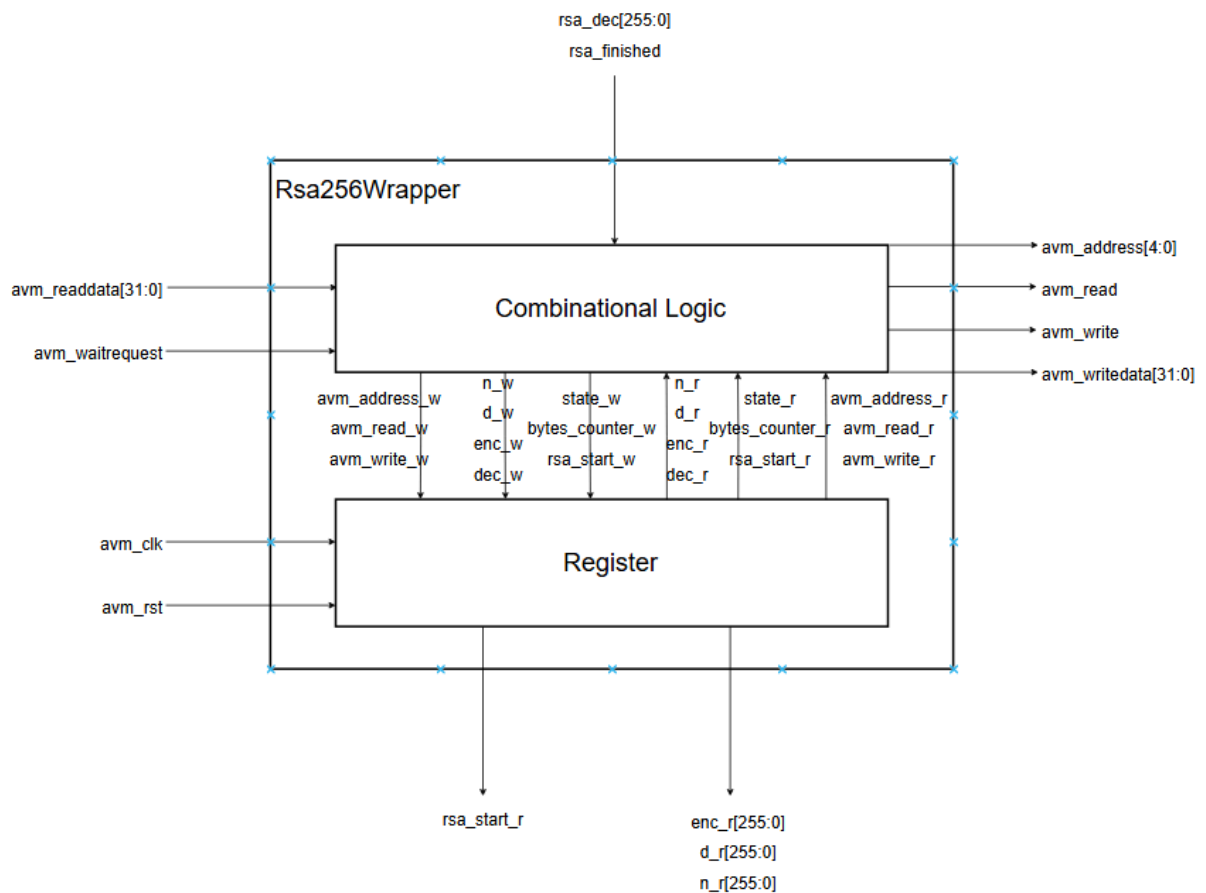
a. Whole system



b. Rsa256Core

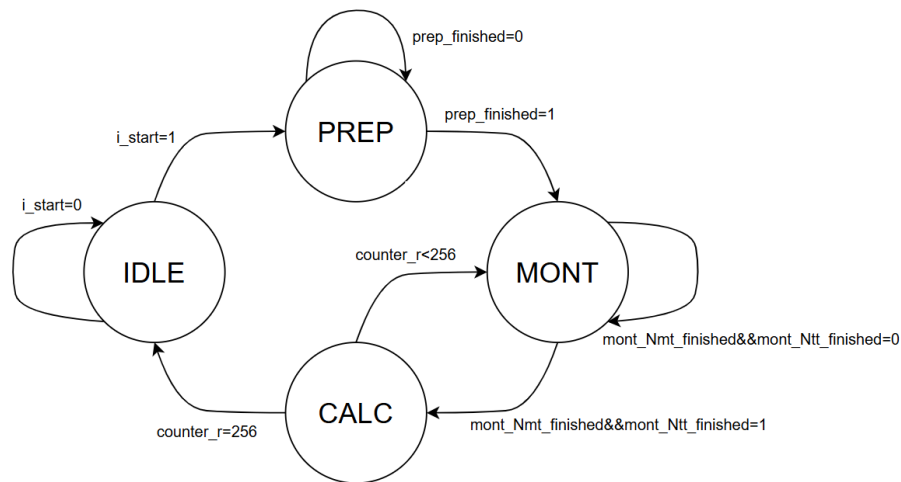


c. Rsa256Wrapper



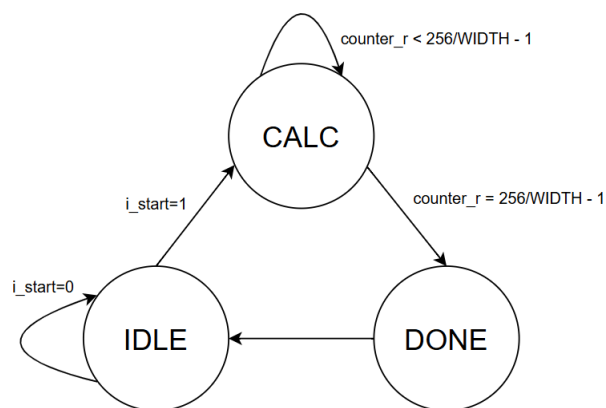
3. State diagram & Explanation

a. Rsa256Core



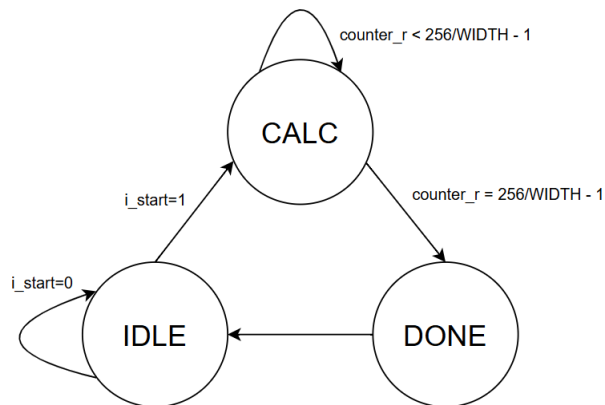
State	Explanation
IDLE	Wait for i_start to start
PREP	Calculate $a \cdot 2^{256} \pmod n$
MONT	Calculate $m \cdot t \cdot 2^{-256} \pmod n$ and $t \cdot t \cdot 2^{-256} \pmod n$ parallelly
CALC	Make sure that the state MONT runs 256 times

b. RsaPrep



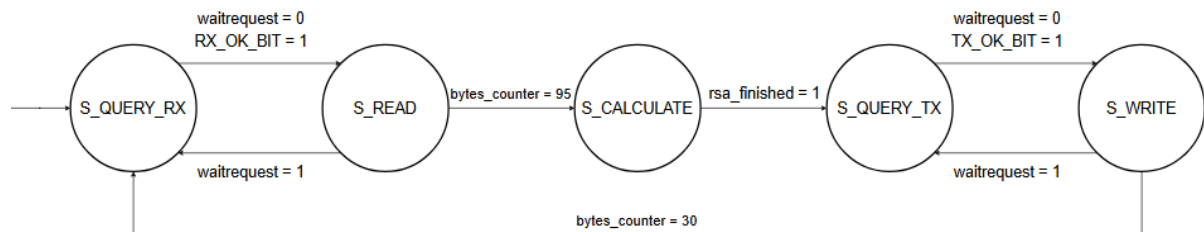
State	Explanation
IDLE	Wait for i_start to start
CALC	Calculate for WIDTH times (see the following pseudocode)
DONE	Return to IDLE

c. RsaMont



State	Explanation
IDLE	Wait for i_start to start
CALC	Calculate for WIDTH times (see the following pseudocode)
DONE	Return to IDLE

d. Wrapper



State	Explanation
S_QUERY_RX	Wait for waitrequest = 0 and register ready to start reading
S_READ	Read data 1 byte at a time, totally read 96 bytes at first (n 32 + d 32 + enc 32) then read 32 bytes for the rest (only enc 32)
S_CALCULATE	Wait for Core to calculate
S_QUERY_TX	Wait for waitrequest = 0 and register ready to start writing
S_WRITE	Write data 1 byte at a time, totally write 31 bytes (dec 31)

4. Algorithm (pseudocode)

a. Rsa256Core

```
1 def Rsa256Core(a, d, n):
2
3     t = RsaPrep(a, n);
4     m = 1;
5
6     FOR i = 0 to 255:
7         if (d[i] == 1): m = RsaMont(m, t, n);
8         t = RsaMont(t, t, n);
9     ENDFOR
10
11     RETURN m;
```

b. RsaPrep

```
1 def RsaPrep(a, n):
2     WIDTH = 32; // can be 2^n; the hardware will limit how large it can be
3     CHUNKS = 256 / WIDTH;
4     y = a;
5
6     FOR i = 0 to CHUNKS-1: // total CHUNKS clk cycles
7         FOR j = 0 to WIDTH-1: // run WIDTH times in each clk cycle
8             y = y << 1;
9             if (y >= n); y = y - n;
10        ENDFOR
11    ENDFOR
12
13    RETURN y;
```

c. RsaMont

```
1 def RsaMont(a, b, n):
2     WIDTH = 32; // can be 2^n; the hardware will limit how large it can be
3     CHUNKS = 256 / WIDTH;
4     m = 0;
5
6     FOR i = 0 to CHUNKS-1: // total CHUNKS clk cycles
7         FOR j = 0 to WIDTH-1: // run WIDTH times in each clk cycle
8             q = (m + a[i*WIDTH+j]*b) mod 2;
9             m = (m + a[i*WIDTH+j]*b + q*n) / 2;
10        ENDFOR
11    ENDFOR
12
13    if (m >= n): m = m - n;
14
15    RETURN m;
```

原本RsaPrep與RsaMont的演算法每次執行需256 cycles，其中每個cycle都執行一次某特定區域電路。但如上圖(b)(c)所示，我們嘗試在單一cycle中連續執行WIDTH次該特定區域電路(事實上是類似電路串聯)，讓總cycle數降為256/WIDTH，意即以更大的電路面積換取更快的RSA解碼速度(前提是累積的delay不能太長，且FPGA有足夠多的硬體資源支援大電路)。但這也會大幅增加編譯時間，當WIDTH=1時，編譯時間約為5分鐘；WIDTH=2時，編譯時間約為7分鐘；WIDTH=8時，編譯時間約為28分鐘；當WIDTH太大時，將無法完成編譯。最終我們採用WIDTH=8，使每次執行RsaPrep與RsaMont所需的cycles由256個降為256/8=32，其中RsaMont的改善帶來的效益最大，總共能減少256*(256-32)=57344個cycles，整體解碼時間約降為原演算法的1/8！

5. Fitter Summary

Quartus II 64-Bit - C:/Users/DCLAB/OneDrive/Lab2/LAB2_TEAM9 - DE2_115

File Edit View Project Assignments Processing Tools Window Help

DE2_115

Project Navigator

- Files
 - rsa_qsys/synthesis/rsa_qsys.v
 - rsa_qsys/synthesis/rsa_qsys.qip
 - src/DE2_115/DE2_115.sv
 - src/DE2_115/DE2_115.sdc
 - src/Rsa256Wrapper.sv
 - src/Rsa256Core.sv

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 - I/O Assignment Warnings
 - Ignored Assignments
 - Incremental Compilation Section
 - Pin-Out File
 - Resource Section
 - I/O Rules Section
 - Device Options
 - Operating Settings and Conditions
 - Messages
 - Suppressed Messages
 - Flow Messages
 - Flow Suppressed Messages
- Assembler
- TimeQuest Timing Analyzer

Flow: Compilation Customize...

Compilation Report - DE2_115

Rsa256

Fitter Summary

Fitter Status	Successful - Tue Oct 07 20:36:04 2025
Quartus II 64-Bit Version	15.0.0 Build 145 04/22/2015 S3 Full Version
Revision Name	DE2_115
Top-level Entity Name	DE2_115
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	6,405 / 114,480 (6 %)
Total combinational functions	5,598 / 114,480 (5 %)
Dedicated logic registers	3,222 / 114,480 (3 %)
Total registers	3222
Total pins	480 / 529 (91 %)
Total virtual pins	0
Total memory bits	0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements	0 / 532 (0 %)
Total PLLs	1 / 4 (25 %)

6. Timing Analyzer

a. Summary

Quartus II 64-Bit - C:/Users/DCLAB/OneDrive/Lab2/LAB2_TEAM9 - DE2_115

File Edit View Project Assignments Processing Tools Window Help

DE2_115

Project Navigator

- Files
 - rsa_qsys/synthesis/rsa_qsys.v
 - rsa_qsys/synthesis/rsa_qsys.qip
 - src/DE2_115/DE2_115.sv
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 - Slow 1200mV 0C Model
 - Fast 1200mV 0C Model
 - Multicorner Timing Analysis Summary
 - Multicorner Datasheet Report Summary

Flow: Compilation Customize...

Compilation Report - DE2_115

TimeQuest Timing Analyzer Summary

Quartus II Version	Version 15.0.0 Build 145 04/22/2015 S3 Full Version
Revision Name	DE2_115
Device Family	Cyclone IV E
Device Name	EP4CE115F29C7
Timing Models	Final
Delay Model	Combined
Rise/Fall Delays	Enabled

b. Setup Summary

Quartus II 64-Bit - C:/Users/DCLAB/OneDrive/Lab2/LAB2_TEAM9 - DE2_115

File Edit View Project Assignments Processing Tools Window Help

DE2_115

Project Navigator

Files

- rsa_qsys/synthesis/rsa_qsys.v
- rsa_qsys/synthesis/rsa_qsys.qip
- src/DE2_115/DE2_115.sv
- src/DE2_115/DE2_115.sdc
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 - Timing Closure Recommendations
 - Setup Summary
 - Hold Summary
 - Recovery Summary

Slow 1200mV 85C Model Setup Summary

	Clock	Slack	End Point TNS
1	my_qsys altpll_0 sd1 pll7 clk[0]	5.940	0.000

c. Hold

Quartus II 64-Bit - C:/Users/DCLAB/OneDrive/Lab2/LAB2_TEAM9 - DE2_115

File Edit View Project Assignments Processing Tools Window Help

DE2_115

Project Navigator

Files

- rsa_qsys/synthesis/rsa_qsys.v
- rsa_qsys/synthesis/rsa_qsys.qip
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 - Recovery Summary
 - Removal Summary
 - Minimum Pulse Width Summary
 - Worst-Case Timing Paths
 - Datasheet Report
 - Metastability Summary

Slow 1200mV 85C Model Hold Summary

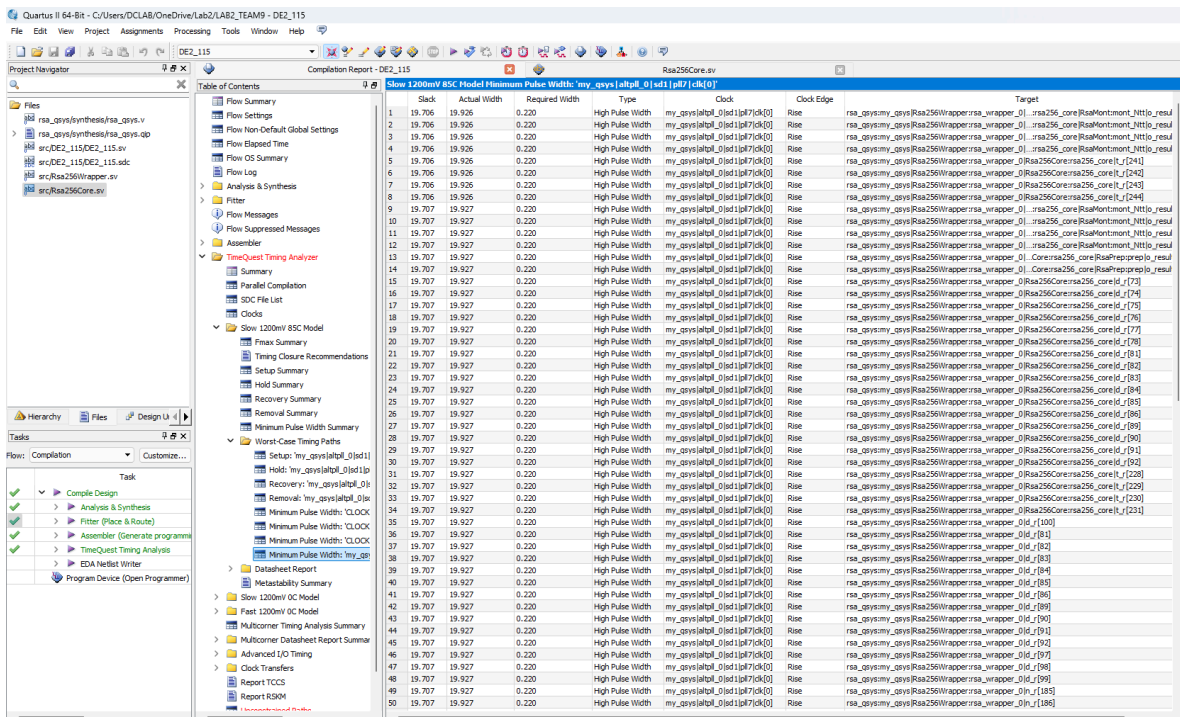
	Clock	Slack	End Point TNS
1	my_qsys altpll_0 sd1 pll7 clk[0]	0.402	0.000

Hierarchy Files Design Utilities

Tasks

Flow: Compilation Customize...

Task



Quartus II 64-Bit - C:/Users/DCLAB/OneDrive/Lab2/LAB2_TEAM9 - DE2_115

File Edit View Project Assignments Processing Tools Window Help

DE2_115

Project Navigator

Files

- rsa_qsys/synthesis/rsa_qsys.v
- rsa_qsys/synthesis/rsa_qsys.qip
- src/DE2_115/DE2_115.sv
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Hierarchy Files Design UI

Tasks

Flow: Compilation Customize...

	Task
✓	Compile Design
✓	> Analysis & Synthesis
✓	> Fitter (Place & Route)
✓	> Assembler (Generate programming file)
✓	> TimeQuest Timing Analysis
	> EDA Netlist Writer
	Program Device (Open Programmer)

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 - Minimum Pulse Width Summary
 - > Worst-Case Timing Paths
 - > Datasheet Report
 - Metastability Summary
 - Multicorner Timing Analysis Summary
 - > Multicorner Datasheet Report Summary
 - > Advanced I/O Timing
 - > Clock Transfers
 - Report TCCS
 - Report RSKM
 - Unconstrained Paths
 - Messages

Compilation Report - DE2_115

Unconstrained Paths

	Property	Setup	Hold
1	Illegal Clocks	0	0
2	Unconstrained Clocks	0	0
3	Unconstrained Input Ports	0	0
4	Unconstrained Input Port Paths	0	0
5	Unconstrained Output Ports	1	1
6	Unconstrained Output Port Paths	1	1

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DE2_115

Project Navigator

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 - Recovery Summary

Fast 1200mV 0C Model Setup Summary

	Clock	Slack	End Point TNS
1	my_qsys altpll_0 sd1 pll7 clk[0]	22.334	0.000

Quartus II 64-Bit - C:/Users/DCLAB/OneDrive/Lab2/LAB2_TEAM9 - DE2_115

File Edit View Project Assignments Processing Tools Window Help

DE2_115

Project Navigator

Files

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Fast 1200mV 0C Model Minimum Pulse Width Summary

	Clock	Slack	End Point TNS
1	CLOCK_50	9.574	0.000
2	CLOCK2_50	16.000	0.000
3	CLOCK3_50	16.000	0.000
4	my_qsys altpll_0 sd1 pll7 clk[0]	19.779	0.000

7. 遇到的問題與心得

這次在[Rsa256Core.sv](#)中的166與258行有用到width來控制串聯的電路數，以一個block要重複執行256次來說，可以寫成每個clk執行width次總共256/width次，因此會做width個一樣電路做串聯，雖然面積會變大並且compile時間會與width

大致呈正比(width = 2 → compile 7mins 14seconds ;width = 8 → compile 27mins 43seconds), 但width數拉高會使我們在做decode時的速度更快, 一開始我們把width設成128, 因此我們compile 3.5hrs還沒好, 後面我們把width降低到1, 即可以在10 mins內完成compile, 最後測試width = 8時仍可以在可接受compile時間內並且提供正確的結果

8. Appendix: Proof of algorithm used in module RsaMont

Assume n is odd and $0 \leq b < n$ (true in RSA cryptosystem)

Let $m_0 = 0$ and $m_{i+1} = \frac{m_i + a[i]b + q_i n}{2}$, where

1. $i = 0, 1, \dots, 255$
2. $a[i]$ is the i -th bit of a
3. $q_i = (m_i + a[i]b) \bmod 2$ s.t. $m_i + a[i]b + q_i n$ is even

First, prove $m_{256} \equiv ab2^{-256} \pmod{n}$

Claim $2^i m_i \equiv (\sum_{j=0}^{i-1} a[j]2^j)b \pmod{n}$ for all $1 \leq i \leq 256$

Prove by mathematical induction:

1. When $i = 1$, $LHS = 2m_1 = a[0]b + q_0 n \equiv a[0]b = RHS \pmod{n}$

2. Suppose it's true for $i = k$, then for $i = k + 1$,

$$\begin{aligned} 2^{k+1}m_{k+1} &= 2^{k+1} \frac{m_k + a[k]b + q_k n}{2} \\ &= 2^k m_k + 2^k a[k]b + 2^k q_k n \\ &\equiv (\sum_{j=0}^{k-1} a[j]2^j)b + 2^k a[k]b \pmod{n} \\ &\equiv (\sum_{j=0}^k a[j]2^j)b \pmod{n} \quad \text{QED.} \end{aligned}$$

Take $i = 256$, then we have

$$2^{256}m_{256} \equiv (\sum_{j=0}^{255} a[j]2^j)b = ab \pmod{n} \dots (1)$$

Since n is odd, $\gcd(2^{256}, n) = 1$

By Bézout's Identity,

$$\exists x, y \in \mathbb{Z} \text{ s.t. } 1 = 2^{256}x + ny \equiv 2^{256}x \pmod{n}$$

$$\text{By (1), } 2^{256}xm_{256} \equiv abx \pmod{n}$$

$$\Rightarrow m_{256} \equiv abx \equiv ab2^{-256} \pmod{n} \quad \text{QED.}$$

Second, prove $m_{256} \in [0, 2n)$

Claim $0 \leq m_i < 2n$ for all $0 \leq i \leq 256$

Prove by mathematical induction:

1. When $i = 0$, $0 \leq 0 = m_0 < 2n$
2. Suppose it's true for $i = k$, then for $i = k + 1$,

$$\begin{aligned} m_{k+1} &\geq 0 \text{ is obvious} \\ m_{k+1} &= \frac{m_k + a[k]b + q_k n}{2} < \frac{2n + n + n}{2} = 2n \quad \text{QED.} \end{aligned}$$

By the first and second proofs, we can conclude that

$$ab2^{-256} \pmod{n} = (m_{256} < n) ? m_{256} : m_{256} - n$$